

Edouard Murat

Gameplay Programmer & Team Lead

St-Leonard (QC) H1R 2X5 • 514-804-5475 • edouardmurat1@gmail.com

Portfolio: edouardmurat.com • GitHub: github.com/StrayEddy

French: fluent • English: fluent

PROFILE

Gameplay programmer and team lead with 20 years of software experience, including five years of production C#. Led a six-person Godot team building curriculum-aligned educational games for Android. Specializes in gameplay systems, AI behavior, procedural generation, shader programming, and reusable game frameworks. Shipped games on Steam, Google Play, and the Apple Store.

TECHNICAL SKILLS

Game Engines	Godot, Unity
Languages	GScript, C#, Python, Java, HTML/CSS/JS, SQL
Shaders & Graphics	Shader scripting (Godot, Unity)
Gameplay Systems	Input (keyboard & mobile), AI behaviors, procedural generation, physics-based mechanics, state machines, combat systems
Tools & Frameworks	Git, GitHub, Blender, VS Code, Django, SQLite, Audacity
Other	Agile workflows, UI integration, performance optimization

EXPERIENCE

Lead Gameplay Programmer (Godot) — Sigma Digital, South Africa (remote) Aug 2025 – Aug 2026

Led a six-person Godot team building curriculum-aligned literacy games delivered on Android tablets to first- and second-grade children in Africa.

- Hired the programming team and led it as technical lead, owning architecture, code review, and sprint planning
- Designed the platform's UI framework and an MVVM + FSM architecture adopted across all activities — decoupling game logic from presentation, enabling per-activity unit testing, and making curriculum content swappable without rebuilding core mechanics
- Delivered six mini-games and supported two rounds of playtesting with the target age group
- Built for low-cost Android tablets, with gameplay mapped to a first- and second-grade reading and writing curriculum
- Worked directly with educators and instructional designers to translate curriculum goals into game mechanics, bridging educational theory and interactive design across each sprint
- Provisioned GPU-enabled AWS EC2 instances to run automated Godot UI tests, enabling headless rendering for CI without local hardware

Gameplay Programmer (Godot) — Shattered Paradise, One More Turn Studios (remote) Jul 2025 – Dec 2025

Sole programmer on a turn-based grid-based dungeon crawler roguelike (Dante's Inferno theme, procedurally generated dungeons). The creative director defined the vision; owned all technical work end-to-end.

- Wrote all Godot gameplay code, shaders, the game website, and backend from scratch
- Prioritized and built features independently across the full stack

Gameplay Programmer (Godot) — Pursuit of Life, Dreamboat (remote) Jun 2025 – Aug 2025

Shader-focused contract on a pixel-art urban life-sim.

- Built a dynamic rain system with customizable impact zones and per-region wetness tracking
- Implemented real-time puddle reflections for sprites, NPCs, and swaying foliage
- Developed day/night lighting driven by a global time-of-day system

Game Developer — EdNoKa, Montreal 2022 – Present

Founder and lead developer of an educational gaming platform blending custom-built quizzes with gameplay across farming, hack & slash, platformer, puzzle, and sports genres.

- Published 5+ games as public Steam playtests using Unity and Godot, integrating real-time quizzes into diverse gameplay loops
- Designed and implemented gameplay systems: player control, AI, procedural generation, level logic, scoring, and feedback
- Implemented unified input systems supporting both keyboard and mobile touch controls
- Created custom shaders for visual storytelling and UX (X-ray vision, mini-cameras, 3D text, underwater effects)
- Built a full-stack Django backend: authentication, data storage, in-game unlockables, and content delivery
- Built and maintained deployment pipelines for Steam, Google Play, and the Apple Store; games live on the public Steam playtest

Web Developer and Graphic Designer — HeadPause, Montreal 2018 – 2020

- Developed and maintained the company website and online presence
- Created animations and visual assets, and ran social advertising campaigns

Java and Web Developer — BMC Software, Montreal

2013 – 2014

- Partial management of the development team in SCRUM, including code reviews and TDD
- Built a new adaptive website to client requirements alongside parallel remote teams
- Interviewed candidates for web developer positions

C# Developer and Analyst — Fujitsu Consulting, Montreal

2008 – 2013

- Five years of production C#: a payment solution for HydroSherbrooke and the daycare management system for the Ministère de la Famille du Québec
- Analyzed client needs, resolved blocking issues, and trained new developers across a three-year project

SELECTED GAME PROJECTS

Full list of playable games at edouardmurat.com

Base of 30 (Godot) — Rogue hack & slash through procedurally generated rooms

2025

- Designed and implemented all core gameplay systems, including combat and player progression
- Developed procedural generation for room layouts, obstacle placement, and navigation paths
- Built custom behavior trees for enemy types with unique attack patterns and movement logic

Aerial Ambush (Godot) — Skill-based platformer with homing missiles

2025

- Implemented responsive, high-speed player movement with precise collision handling
- Developed homing missile AI with adjustable targeting and physics-based steering
- Designed and balanced dodge-based gameplay loops for short sessions and high replayability

Agent Ka (Godot) — Waldo-like game in a generated city with crowds and powers

2024

- Built a procedural city generation system with randomized buildings, roads, and crowds
- Programmed crowd simulation with dynamic movement, behaviors, and density-based logic
- Implemented X-ray, mini-camera, and slow-motion powers via custom shaders

Veggies Island (Godot) — Farming game on a cartoonish island

2024

- Implemented the core farming loop: planting, watering, harvesting, and crop growth stages
- Built procedural island generation with randomized terrain, resources, and farm zones
- Developed a lightweight inventory system for harvested crops and seeds

Pico Golf (Godot) — Mini golf challenges with physics-based terrain

2023

- Designed 3D mini-golf courses with increasing difficulty and varied sea-floor layouts
- Integrated animated fish and custom water and caustics shaders to enhance immersion
- Developed audio-visual feedback systems for a calming, low-pressure experience

Tomb (Godot) — Doom-inspired shooter for Godot Wild Jam #42

2022

- Built from scratch in under one week during a jam, demonstrating rapid prototyping
- Engineered arcade-style shooter feel: player movement, shooting, and health systems
- Designed enemy AI using state machines and pathfinding logic with trap systems

Earlier titles include Left 4 Dead 2D, Meditation Card Game, Fruit, Archer, and Meditation World VR.

EDUCATION

B.A. Psychology — Concordia University, Montreal

2015 – 2017

B.Eng. Computer Engineering — University of Sherbrooke

2005 – 2009

College Diploma, DEC in Sciences — Stanislas College, Montreal

2003 – 2005

CERTIFICATIONS

Certified Associate Python Programmer — Python Institute

2024

Oracle Professional Certification, Java SE 7 Programmer

2013

Oracle Associate Certification, Java SE 7 Programmer

2012

References available upon request.